Graph Database Cloud Benchmarking

Benchmark CognoDB Cloud against other managed graph database platforms and publish a reproducible, honest comparison.

Deliverable: A public GitHub repository URL

Deadline: 48 hours from receiving this assignment

Submit to: hr@wexa.ai

1. About the Role and Why ?

Wexa AI is a deep-tech AI lab building infrastructure components necessary for running AI both on software and on hardware. This role requires technology evangelism and passion behind disseminating an innovation done in the laboratory. It will be an internal role with a public facing evangelism for community building. Our way of measuring is how you can explain a deep technical subject and reach an audience. Please be creative in your approaches.

2. Objective

Design and run a benchmark suite that compares CognoDB.com with other managed graph database cloud platforms on the same dataset and the same workloads. We are evaluating your engineering rigor: fair methodology, reproducible automation, and clear, honest reporting — not whether any particular database "wins".

3. Set up CognoDB Cloud

1. **Create an account.** Go to https://console.cognodb.com/signup and sign up. The free tier requires no credit card.
2. **Create a free instance.** From the console, create a free (c0) instance and pick a region. It provisions in under a minute. Each workspace gets one free instance.
3. **Save your connection details.** You will get a connection URI of the form bolt+s://<instance-id>.databases.cognodb.cloud and a generated password for the user “cognodb”. The password is shown exactly once — copy or download it immediately and store it where your code reads its secrets.
4. **Connect with an official Neo4j driver.** Install the official Neo4j driver for your language, point it at your bolt+s:// URI with username “cognodb” and your saved password, and run your first Cypher query. No other code changes are needed.

Note on fairness — same resources everywhere

Run every database on the same resources: the same (or as close as the tiers allow) vCPU, RAM and storage allocation for every platform, so no database gets a hardware advantage. The CognoDB free tier is intentionally small (burstable 0.5 vCPU, 256 MB RAM, 1 GB disk) — use the free or entry tier of every platform you compare, record each platform's advertised specs in your README, and size your dataset so it fits the smallest tier. Comparing databases on unequal resources (e.g. a free tier against a paid tier) is a methodology error.

4. Databases to compare

Benchmark CognoDB Cloud against at least four other graph databases — which ones is entirely your choice. Choosing credible, comparable databases is itself part of the evaluation. Every database must run under the same resource limits (see the fairness note above): equivalent vCPU, RAM and storage for all of them, with each one's specs documented in your README.

Free tiers, free trials or self-hosted deployments capped to the same resources are all fine.

5. Benchmark requirements

5.1 Dataset

- Use a public dataset with at least 100,000 relationships — for example a sample of the SNAP soc-Pokec social network, a movie/actor graph, or a citation network. State the exact source and size (node count, relationship count) in the README.
- Load the identical dataset into every platform, and document the load method you used for each (driver batching, bulk import tool, CSV import, etc.).
- Keep the dataset small enough to fit every platform's free tier — roughly 100k–500k relationships is a good range.

5.2 Required metrics

Measure every metric below on every platform. Run each query workload enough times to report stable numbers (we suggest ≥ 100 iterations per read workload after warm-up), and report the percentiles — not just averages.

Category	Metric	What to report
Data loading	Ingest throughput	Nodes/second and relationships/second, plus total wall-clock load time
Traversals	1-hop, 2-hop and 3-hop query latency	p50 and p95 latency (ms) for each hop depth, from a randomly chosen set of start nodes
Lookups	Point lookup and indexed/filtered lookup	p50 and p95 latency (ms); state which properties are indexed on each platform
Aggregations	Count / group-by style query	p50 and p95 latency (ms) for at least one aggregation over a label or relationship type
Mixed workload	Concurrent read/write throughput	Sustained queries/second with a stated client concurrency (e.g. 10–40 concurrent clients) and read/write mix
Footprint	Resource usage where observable	Whatever the platform exposes: stored data size, memory usage, instance specs. Say “not observable” where it is not.

5.3 Methodology rules

- Same resources for every database — equivalent vCPU, RAM and storage on every platform, documented in the README.
- Same dataset, same logical queries, same client machine and region for every platform.
- Warm up each database before measuring; report cold-start numbers separately if you include them.
- Automate everything — a single script (or one per platform) should load the data, run the workloads and emit the results.
- Record every caveat honestly: free-tier throttling, network variance, query-language differences, timeouts, failed runs. Honest caveats earn credit; hidden ones lose it.

6. Deliverables

A GitHub repository containing:

- **Benchmark code** — data loaders, workload runners and any harness/orchestration code.
- **Reproducible instructions** — anyone with free-tier accounts should be able to re-run your benchmarks from the README alone.
- **README with the full results matrix** — every metric from section 5.2 for every platform tested, in clearly formatted tables (charts are a plus), plus your methodology, environment/instance specs, dataset details and caveats.
- **Analysis** — a short written section explaining what the numbers show and, where you can, why the platforms differ.
  If you wish to keep the github repo private, please give us necessary access to access the repo.

7. What a strong submission looks like

At minimum, cover CognoDB plus four other graph databases of your choice with every required metric measured under the same resource limits, fully scripted runs and clear README tables. Beyond that, submissions stand out through:

- A well-argued selection of databases and a fair, well-documented resource setup.
- Concurrency sweeps on the mixed workload (e.g. 1 / 10 / 40 clients).
- Charts of the results and separation of warm vs. cold numbers.
- Rigorous methodology: variance across repeated runs and a fairness analysis of free-tier limits.
- Root-cause reasoning about why the platforms differ, and a harness others could extend.
- A genuinely engaging, well-written post or article that makes the methodology and findings understandable to a broad technical audience — and that earns real engagement (stars, reactions, views), not just publication.

8. Evaluation criteria

Criterion	Weight	What we look for
Methodology & fairness	25%	Same data/queries everywhere, sensible warm-up, tier parity, honest caveats
Completeness of metrics	20%	All required metrics measured and reported for every platform
Reproducibility & code quality	20%	One-command runs, clean readable code, pinned dependencies
README & analysis	15%	Clear results matrix, well-structured writing, insight into the numbers
communication	20%	Quality and clarity of the report per section 1

9. Submission

- Make the GitHub repository.
- Email both the repository URL to hr@wexa.ai with the subject line “CognoDB Assignment 1 – <Your Name>”.
- Submit within 48 hours of receiving this assignment.

- Do not include your CognoDB (or any platform) passwords or connection URIs in the repository — read them from environment variables.

10. Questions & support

If you hit a problem with CognoDB Cloud itself (signup, provisioning, connectivity), email cognodb@wexa.ai. For questions about the assignment, reply to the email that delivered it. Use of AI coding assistants is fine — you must be able to explain and defend every part of your submission in the follow-up interview.